1 Microsoft SAP to Fabric Connectivity Architecture Patterns (2026)

April 2026 – Complete Reference Guide

Based on: Microsoft Fabric Documentation, Ignite 2025, FabCon 2026

1.1 Overview

Microsoft Fabric offers multiple ways to connect to SAP systems. These range from data-movement patterns (batch ETL, CDC, Mirroring) to federation patterns where SAP data remains in place and is queried live, as well as event-driven patterns for operational analytics. The right approach depends on freshness requirements, data governance ownership, SAP source system, availability of SAP Datasphere, and whether data movement into OneLake is acceptable or required.

This document describes the primary architectural patterns available in 2026 for integrating SAP systems with Microsoft Fabric. These patterns range from full data ingestion into OneLake to semantic federation and event-driven operational analytics, enabling organizations to select the appropriate integration model based on governance, licensing, and performance constraints.

Power BI Direct Lake (GA March 2026): Data ingested into OneLake – whether via connectors, Copy Job, or Mirroring – can be consumed by Power BI using Direct Lake mode. This eliminates the need for data import or intermediate semantic models, ensuring dashboards reflect the latest data with near-in-memory performance.

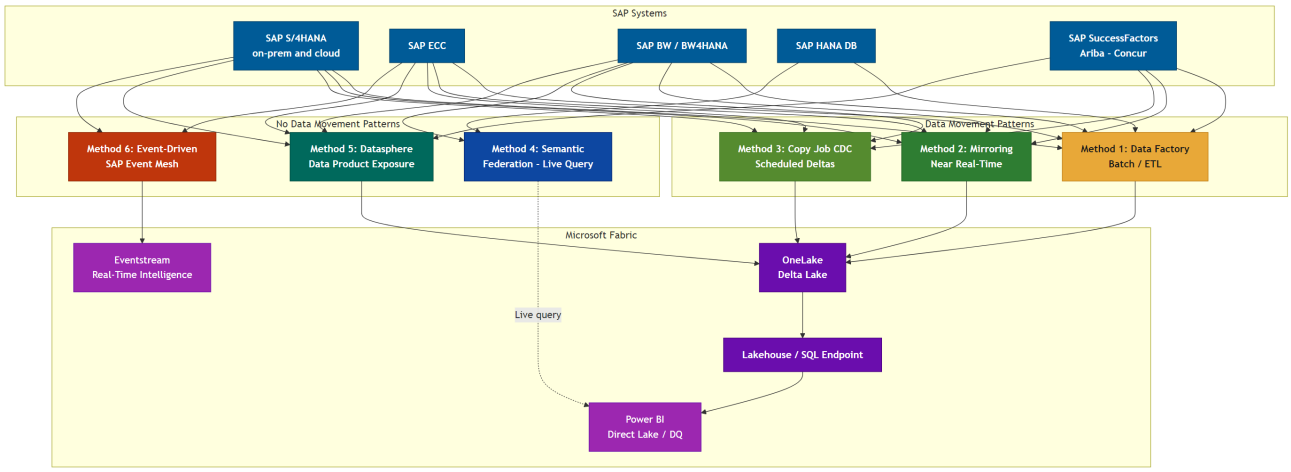

Figure 1: Diagram 1

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1.2 Method 1 – Data Factory Connectors (Batch / ETL)

Seven dedicated SAP connectors are available in Microsoft Fabric Data Factory for scheduled or on-demand data extraction. An **OData connector** is also available for SAP SaaS applications (SuccessFactors, S/4HANA Cloud, C4C) that expose OData APIs.

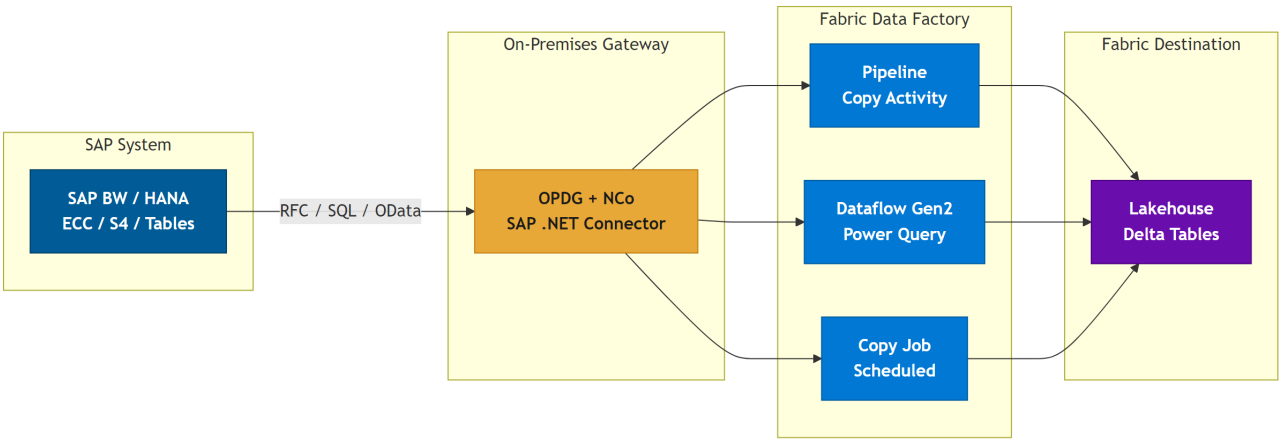

Figure 2: Diagram 2

1.2.1 Connector Reference Table

Connector	Dataflow Gen2	Pipeline (Copy)	Copy Job	Gateway
SAP BW Application Server	✓ Import + DQ	✗	✗	OPDG + NCo 3.x
SAP BW Message Server	✓ Import + DQ	✗	✗	OPDG + NCo 3.x
SAP BW Open Hub – App Server	✓	✓	✗	OPDG
SAP BW Open Hub – Msg Server	✓	✓	✗	OPDG
SAP HANA Database	✓ incl. DQ	✓ Lookup + Copy	✓	OPDG
SAP Table – App Server	✗	✓	✓	OPDG + NCo
SAP Table – Msg Server	✗	✓	✗	OPDG
OData (generic)	✓	✓	✗	None / OPDG

Legend: ✓ = Supported | ✗ = Not supported | DQ = DirectQuery

1.2.2 When to Use

- **SAP BW connectors** – BW InfoProviders, BEx queries, Open Hub destinations. BW 7.3, 7.5, BW/4HANA 2.0. Best for aggregated datasets.
- **SAP HANA** – HANA views, tables, stored procedures. Also supports DirectQuery via Dataflow Gen2.
- **SAP Table** – ABAP table/view extraction via RFC (`VBAK`, `MARA`, `KNA1`). Medium-volume extractions.
- **OData** – SAP apps exposing OData APIs (SuccessFactors, S/4HANA Cloud, C4C). Small-to-medium volumes.

1.2.3 Infrastructure Prerequisites

All SAP connectors (except OData) require this infrastructure, **even when SAP runs in the cloud**:

1. **On-Premises Data Gateway (OPDG)** deployed near the SAP system
2. **SAP .NET Connector (NCo) 3.0 or 3.1** installed on the gateway server
3. **Network access** to SAP: RFC ports (33XX), HANA port (30015), via VPN/VNet or ExpressRoute
4. **SAP technical account** with appropriate authorizations (`S_RFC`, `S_TABU_DIS`)

Important limitations:

- **No Fabric-managed CDC** – relies on SAP-native delta mechanisms such as ODP extractors or BW Open Hub delta queues. Fabric itself does not track SAP-side changes; you must manage incremental logic through SAP-provided capabilities.
- **No dedicated connector for SAP SaaS** – SuccessFactors, Ariba, Concur require OData or Mirroring/CDC.
- **Performance impact on SAP** – large RFC/BW extractions consume SAP resources. For massive tables, prefer HANA direct or Mirroring.

1.3 Method 2 – Mirroring for SAP (Near Real-Time)

Mirroring for SAP provides **continuous, near real-time replication** of SAP data into OneLake, without any custom ETL. It operates as a Fabric item ("mirrored database"), fully managed by the platform.

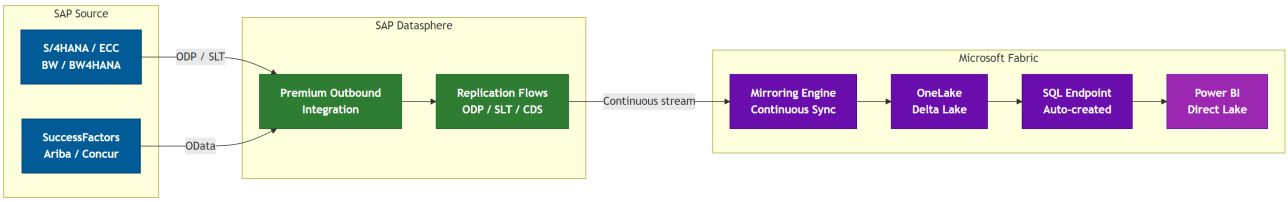

Figure 3: Diagram 3

1.3.1 Supported SAP Sources

SAP System	Deployment	Supported
SAP S/4HANA	On-premises	✓
SAP S/4HANA Cloud	Cloud (public + private)	✓
SAP ECC	On-premises	✓
SAP BW	On-premises	✓
SAP BW/4HANA	On-premises and cloud	✓
SAP SuccessFactors	SaaS	✓
SAP Ariba	SaaS	✓
SAP Concur	SaaS	✓

Other SAP systems (CRM, SRM, SCM) based on NetWeaver ABAP are covered via ODP/SLT, same as ECC. Any SAP system supporting ODP extraction is eligible.

1.3.2 Key Benefits

- **Zero ETL** – schema evolution handled automatically; raw replication (transforms applied downstream)
- **Near real-time** – latency typically seconds to a few minutes
- **End-to-end lineage** – full governance and audit trail
- **Native Fabric integration** – SQL endpoint, Power BI Direct Lake, Notebooks, Lakehouses
- **Limited impact on SAP when SLT-based ODP extraction is already operational** – if SLT triggers and ODP queues are pre-configured, Mirroring adds marginal load. However, initial SLT configuration introduces logging overhead on source tables (trigger installation, change-log table creation). Plan the initial setup during low-activity windows.
- **Up to 1,000 tables** per mirrored database (increased from ~100 at FabCon 2026)

1.3.3 Prerequisites

1. **SAP Datasphere** with **Premium Outbound Integration** (mandatory)
2. Replication Flows configured in Datasphere
3. **On-premises SAP:** Data Provisioning Agent or SAP Cloud Connector on-site
4. **SAP Cloud sources:** OData connections activated and registered in Datasphere
5. Fabric capacity (F2+ recommended)
6. Network: SAP Datasphere to Fabric (outbound HTTPS)

SAP licensing note: SAP requires official extraction products (Datasphere, Data Intelligence) for ODP-based extraction. This reinforces Mirroring as the strategic SAP-endorsed path.

1.4 Method 3 – Copy Job CDC for SAP

Introduced at **Ignite 2025**, Copy Job supports **Change Data Capture (CDC)** for SAP via Datasphere. Unlike Mirroring (autonomous), Copy Job CDC provides **explicit orchestration control** within a Data

Factory pipeline.

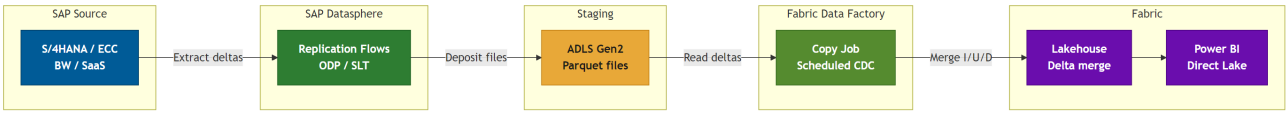

Figure 4: Diagram 4

1.4.1 How It Works

Two-stage mechanism:

1. **SAP Datasphere** extracts initial data then delta changes (ODP/SLT) and deposits Parquet files on **Azure Data Lake Storage Gen2**.
2. **Fabric Copy Job** reads those files and merges inserts/updates/deletes into the Fabric Lakehouse (Delta).

1.4.2 Feature Summary

Feature	Value
Change types captured	Inserts, Updates, Deletes
Watermark column needed	✗ Not required
Manual refresh needed	✗ Scheduled trigger
Merge destination	Lakehouse (Delta)
Intermediate storage	ADLS Gen2 / S3 / GCS

1.4.3 Prerequisites

1. **SAP Datasphere** with **Premium Outbound Integration** (same as Mirroring)
2. Data Provisioning Agent for on-premises sources
3. Replication Flows targeting a cloud storage container (ADLS Gen2)
4. Fabric Copy Job configured to read from that container

1.4.4 When to Prefer over Mirroring

- **Control the schedule** (e.g., every 15 min during business hours, pause at night)
- **Multi-source pipeline** with additional transforms, validations, or joins
- **Limit continuous load** – scheduled bursts vs. 24/7 streaming
- **Monitoring** is split: Datasphere (replication health) + Fabric (Copy Job runs)

> **Latency** depends on scheduled frequency. A 5-minute interval = up to 5 minutes stale. For continuous freshness, use Mirroring.

1.4.5 Copy Job Optimizations (FabCon 2026)

- **Auto-partitioning** for better performance on large copies
- **Automatic audit columns** for load tracking
- **Zero CU cost** when no data changes exist

1.5 Method 4 – Semantic Federation (No Data Movement)

Microsoft Fabric semantic models (via the Power BI engine) can connect **live** to SAP semantic layers without replicating any data into OneLake. This is **federation, not ingestion** – SAP remains the system of record and the query engine.

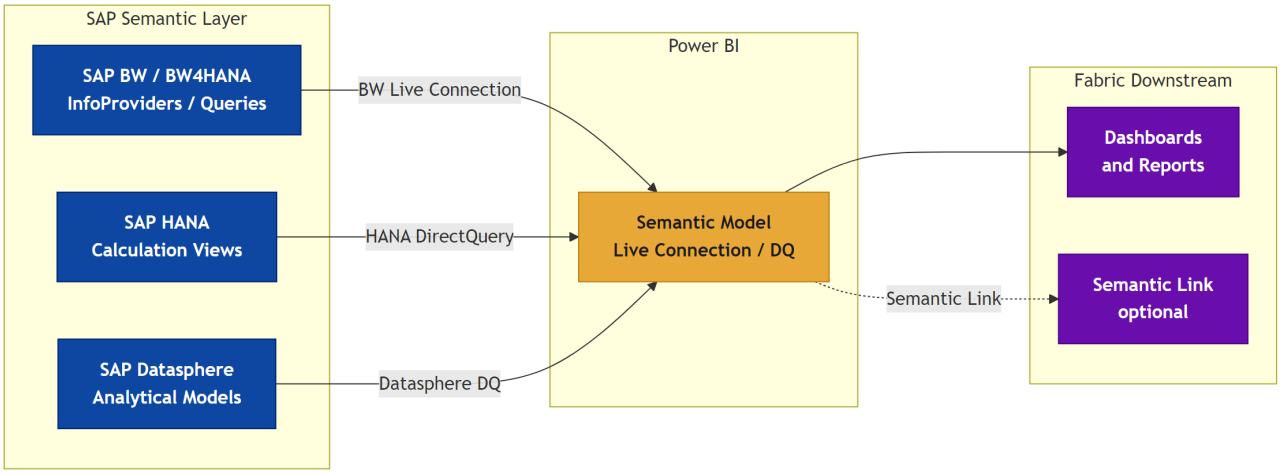

Figure 5: Diagram 5

1.5.1 Connection Types

Connection	SAP Source	Protocol	Data Movement
BW Live Connection	SAP BW / BW4HANA	MDX via OPDG	✗ None
HANA DirectQuery	SAP HANA Calculation Views	SQL via OPDG	✗ None
Datasphere DirectQuery	Datasphere Analytical Models	OData / SQL	✗ None

1.5.2 Key Benefits

- **Zero data replication** – no SAP data enters OneLake; all queries are executed on SAP infrastructure
- **SAP-governed security enforcement** – row-level security, authorizations, and data classifications defined in SAP are enforced at query time
- **SAP business logic reuse** – CDS views, BW hierarchies, currency conversions, and calculated key figures are executed on the SAP engine, not re-implemented in Fabric
- **Regulatory compliance** – suitable for regulated industries (banking, pharma, public sector) where data residency or extraction restrictions apply
- **Hybrid virtualization** – Power BI semantic models can combine live SAP connections with imported data from other sources in a single report

1.5.3 Limitations

- **Query performance depends on SAP infrastructure** – response time is bounded by SAP system capacity and network latency

- **No offline access** – if SAP is unavailable, dashboards are unavailable
- **Requires On-Premises Data Gateway** for BW and HANA connections
- **Not suitable for large-scale data science / notebook workloads** – federation is optimized for BI consumption, not for Spark-based analytics

1.5.4 When to Use

- Data must remain in SAP for regulatory or contractual reasons
- SAP business logic (hierarchies, currencies, authorizations) must be enforced at the source
- Dashboard consumption only – no downstream Spark/notebook processing needed
- SAP infrastructure has sufficient capacity to handle concurrent BI queries

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1.6 Method 5 – Datasphere-Mediated Data Product Exposure

In this architecture, **SAP Datasphere acts as a governed data product layer**. Rather than Fabric extracting data directly from SAP operational systems, Datasphere curates, governs, and exposes analytical data products that Fabric consumes.

Figure 6: Diagram 6

1.6.1 How It Works

1. SAP operational systems replicate data into **SAP Datasphere** using standard mechanisms (ODP, SLT, CDS)
2. Datasphere applies **transformations, business rules, and governance** to produce curated analytical models (data products)
3. Datasphere exports these data products to a cloud storage layer (ADLS Gen2, S3), typically using columnar analytical formats such as Parquet
4. Fabric accesses this storage via **OneLake Shortcuts** – no additional copy into OneLake
5. Power BI consumes the data through **Direct Lake** mode

1.6.2 Impact on Architecture

Aspect	Implication
Data ownership	SAP team owns the data product definition and quality; Fabric team consumes
Extraction licensing	Compliant with SAP licensing – extraction is performed by SAP Datasphere (an SAP product), not by a third-party tool
Governance boundary	Data governance and business rules are enforced in Datasphere before data leaves SAP's perimeter
SAP strategic direction	Aligns with SAP's recommended architecture where Datasphere is the authorized data sharing layer

1.6.3 When to Use

- Organization has invested in SAP Datasphere as a strategic data platform
- SAP data team produces governed data products consumed by multiple downstream platforms (not just Fabric)
- Strict data governance requires that business logic and quality rules are applied before data leaves SAP
- SAP extraction licensing mandates that only SAP-approved tools perform outbound data replication

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1.7 Method 6 – Event-Driven Integration (Operational Analytics)

SAP business events can flow into Microsoft Fabric for **real-time operational analytics** without bulk data extraction. This pattern targets individual business transactions, not full tables.

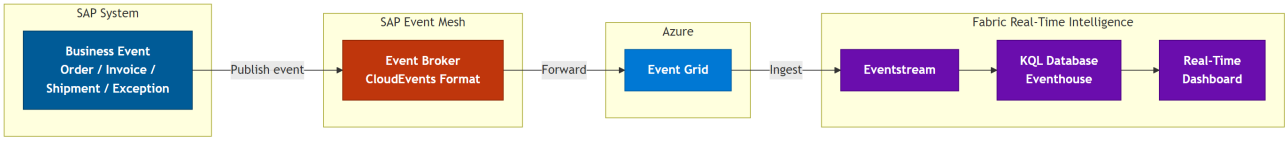

Figure 7: Diagram 7

1.7.1 Supported Event Patterns

SAP Event Source	Event Examples	Latency
S/4HANA Business Events	Order created, Invoice posted, Delivery shipped	Sub-second
SAP ECC (via BTP)	Goods receipt, Production order exception	Seconds
SAP Integration Suite	Composite events from multiple SAP systems	Seconds

1.7.2 Architecture Components

- **SAP Event Mesh** (part of SAP BTP) – publishes business events in CloudEvents format
- **Azure Event Grid** – receives events from SAP Event Mesh and routes to Fabric
- **Fabric Eventstream** – ingests, transforms, and routes events within Fabric
- **KQL Database / Eventhouse** – stores event data for time-series analysis and anomaly detection
- **Real-Time Dashboards** – visualize operational KPIs with sub-minute refresh

1.7.3 When to Use

- Operational monitoring: detect exceptions, SLA breaches, or anomalies in SAP business processes
- Event-triggered automation: SAP order → Fabric enrichment → downstream action
- Real-time KPIs: live order-to-cash metrics, production throughput, logistics tracking
- Complement to bulk integration – events for freshness, Mirroring/CDC for completeness

This pattern supports operational analytics, not bulk historical ingestion. For complete datasets (all orders, all materials), use Methods 1-3 or 5. Event-driven integration captures individual business transactions as they occur.

1.7.4 Prerequisites

1. **SAP BTP** subscription with **SAP Event Mesh** enabled
2. SAP business events activated in S/4HANA or ECC (requires SAP Basis configuration)
3. Azure Event Grid subscription configured to receive from SAP Event Mesh
4. Fabric capacity with Real-Time Intelligence workload enabled

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1.8 Alternative Approaches

1.8.1 OneLake Shortcuts

If SAP data already exists in external storage (ADLS, S3), create a **OneLake Shortcut** to make it available in Fabric without re-copying.

1.8.2 Third-Party ETL Tools

Informatica, Boomi, Theobald, and others offer SAP connectors writing to OneLake. Microsoft's strategic direction favors native connectors and SAP Datasphere.

1.8.3 Logic Apps / Power Automate

For event-driven micro-integrations (e.g., SAP order creation triggers a Fabric action). Not suitable for bulk data.

1.9 Decision Guide

1.10 Key Announcements

1.10.1 Ignite 2025 – November 2025

Feature	Status	Coverage
Mirroring for SAP	◑ Preview	S/4HANA, BW, BW/4HANA, SuccessFactors, Ariba
Copy Job CDC for SAP	✓ GA	SAP via Datasphere to Lakehouse

1.10.2 FabCon 2026 – March 2026

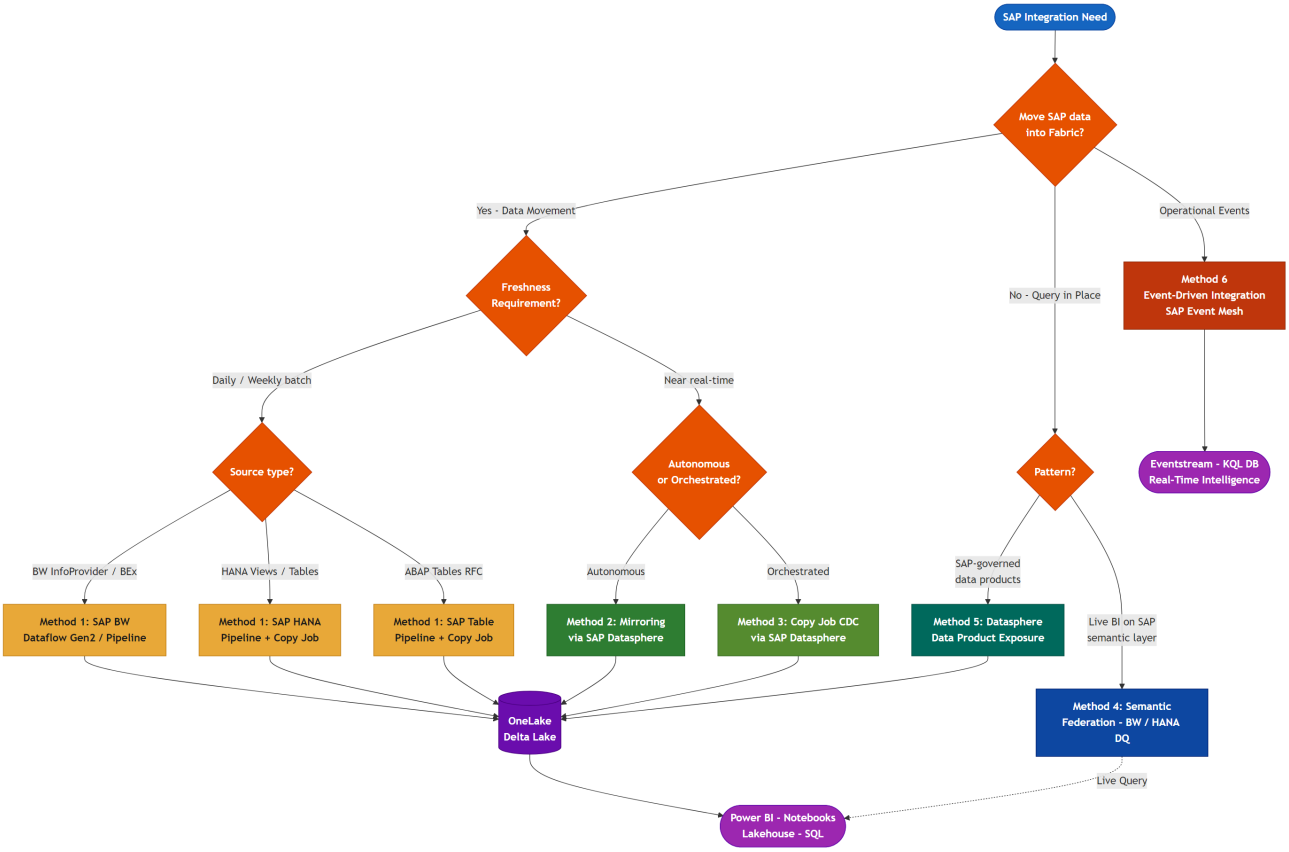

Figure 8: Diagram 8

Feature	Status	What's New
Mirroring for SAP	✓ GA	+ SAP ECC, + Concur. Up to 1,000 tables.
Copy Job enhancements	✓ GA	Auto-partitioning, audit columns, zero-cost
Direct Lake for Power BI	✓ GA	Dashboards read Delta directly from OneLake

Docs: Microsoft Fabric Mirrored Databases From SAP

1.11 Comparison Summary

Criteria	Batch Connectors	Copy Job CDC	Mirroring	Semantic Federation	Datasphere Products	Event-Driven
Data movement	✓ To OneLake	✓ To OneLake	✓ To OneLake	✗ None	✓ To storage	✗ Events only
Freshness	Hourly to daily	Minutes	Near real-time	Live query	Export schedule	Sub-second
Custom ETL needed	Required	Minimal	None	None	Datasphere-side	Event routing
SAP Datasphere	✗ Not required	✓ Required	✓ Required	✗ Optional	✓ Required	✗ Not required
SAP BTP	✗	✗	✗	✗	✗	✓ Required
Native CDC	✗ SAP-side only	✓ Scheduled	✓ Continuous	N/A	N/A	N/A (events)
SAP sources	BW, HANA, Tables	Full landscape	Full landscape	BW, HANA, DSphere	Full landscape	S/4, ECC
Power BI access	DQ + Import	Direct Lake	SQL EP + Direct Lake	Live Connection / DQ	Direct Lake	RT Dashboard

Criteria	Batch Connectors	Copy Job CDC	Mirroring	Semantic Federation	Datasphere Products	Event-Driven
Governance owner	Fabric team	Fabric team	Fabric team	SAP team	SAP team	SAP (event ownership) / Fabric (analytics ownership)
Use case	Analytical	Analytical	Analytical	Analytical (BI)	Analytical	Operational
GA status	All GA (2023)	GA (Nov 2025)	GA (Mar 2026)	GA	GA	GA (components)

Legend: ✓ Yes / Required | ✗ No / Not required | N/A = Not applicable

1.12 Recommendations by Scenario

Historical bulk load (migrating years of data): Use **Method 1 – batch connectors** (SAP HANA or SAP Table via Pipeline Copy job with partitioning).

Regular analytics refresh (daily/hourly dashboards): Use **Method 1** for simple cases, or **Method 3 – Copy Job CDC** for incremental deltas.

Real-time operational analytics (live sales, inventory): Use **Method 2 – Mirroring** + Power BI Direct Lake for continuous dashboard freshness.

SAP SaaS without Datasphere (SuccessFactors, Ariba): Use the **OData connector** (Method 1) for moderate volumes; invest in Datasphere for scale.

Multi-source orchestrated pipeline (SAP + other sources): Use **Method 3 – Copy Job CDC** in a Data Factory pipeline with transforms and validations.

Data must stay in SAP (regulatory, contractual, or governance constraints): Use **Method 4 – Semantic Federation** for BI consumption via Power BI Live Connection / DirectQuery.

SAP team owns data products (governed exposure model): Use **Method 5 – Datasphere Data Products** exported to cloud storage, consumed via OneLake Shortcuts.

Real-time business event monitoring (order tracking, SLA alerting): Use **Method 6 – Event-Driven** via SAP Event Mesh + Azure Event Grid + Fabric Eventstream.

No SAP Datasphere available: Use **Method 1 – batch connectors** + OPDG. For BI without data movement, consider **Method 4 – Semantic Federation** via BW Live Connection or HANA DirectQuery.

1.13 Appendix A – References

1.13.1 Mirroring for SAP

Resource	Link
Mirrored Databases from SAP	https://learn.microsoft.com/fabric/mirroring/sap
Mirroring Overview	https://learn.microsoft.com/fabric/mirroring/overview
Extended Capabilities (CDF, Views)	https://learn.microsoft.com/fabric/mirroring/extended-capabilities
Troubleshooting Guide	https://learn.microsoft.com/fabric/mirroring/troubleshooting

1.13.2 Data Factory SAP Connectors

Resource	Link
Connector Overview (all)	https://learn.microsoft.com/fabric/data-factory/connector-overview
SAP BW Open Hub	https://learn.microsoft.com/fabric/data-factory/connector-sap-bw-open-hub-overview
SAP HANA	https://learn.microsoft.com/fabric/data-factory/connector-sap-hana-database-overview
SAP Table	https://learn.microsoft.com/fabric/data-factory/connector-sap-table-overview
SAP BW Application Server	https://learn.microsoft.com/power-query/connectors/sap-bw/application-setup-and-connect
OData Connector	https://learn.microsoft.com/fabric/data-factory/connector-odata-overview

1.13.3 Copy Job and CDC

Resource	Link
What is Copy Job	https://learn.microsoft.com/fabric/data-factory/what-is-copy-job

Resource	Link
CDC in Copy Job	https://learn.microsoft.com/fabric/data-factory/copy-job-change-data-capture
Copy Job Monitoring	https://learn.microsoft.com/fabric/data-factory/copy-job-workspace-monitoring

1.13.4 Power BI and OneLake

Resource	Link
Direct Lake Mode	https://learn.microsoft.com/fabric/fundamentals/direct-lake-overview
OneLake Shortcuts	https://learn.microsoft.com/fabric/onelake/onelake-shortcuts

1.13.5 Announcements

Resource	Link
Ignite 2025 Feature Summary	https://blog.fabric.microsoft.com/en-us/blog/fabric-november-2025-feature-summary
FabCon 2026 Feature Summary	https://blog.fabric.microsoft.com/en-us/blog/fabric-march-2026-feature-summary
FabCon 2026 Hero Blog	https://aka.ms/FabCon-SQLCon-2026-news
SAP in Fabric Blog (Sept 2024)	https://blog.fabric.microsoft.com/en-us/blog/connecting-to-sap-data-in-microsoft-fabric

1.13.6 SAP Datasphere

Resource	Link
SAP Datasphere Docs	https://help.sap.com/docs/SAP_DATASPHERE
Premium Outbound Integration	https://help.sap.com/docs/SAP_DATASPHERE/be5967d099974c69b77f4549425ca4c0/eb7ff31

Resource	Link
Data Provisioning Agent	https://help.sap.com/docs/SAP_DATASPHERE/935116dd7c324355803d4b85809cec97

1.13.7 Infrastructure

Resource	Link
On-Premises Data Gateway	https://learn.microsoft.com/data-integration/gateway/service-gateway-onprem
SAP .NET Connector (NCo)	https://support.sap.com/en/product/connectors/msnet.html

1.13.8 Semantic Federation and Live Connections

Resource	Link
SAP BW Connector (Power Query)	https://learn.microsoft.com/power-query/connectors/sap-bw/application-setup-and-connect
SAP HANA DirectQuery in Power BI	https://learn.microsoft.com/power-bi/connect-data/desktop-directquery-sap-hana
Power BI Live Connection Overview	https://learn.microsoft.com/power-bi/connect-data/desktop-directquery-about

1.13.9 Event-Driven Integration

Resource	Link
SAP Event Mesh	https://help.sap.com/docs/SAP_EM
Azure Event Grid	https://learn.microsoft.com/azure/event-grid/overview
Fabric Eventstream	https://learn.microsoft.com/fabric/real-time-intelligence/event-streams/overview
Fabric Real-Time Intelligence	https://learn.microsoft.com/fabric/real-time-intelligence/overview

1.14 Appendix B – Glossary

Acronym	Definition
CDC	Change Data Capture – track inserts, updates, and deletes
CDS	Core Data Services – SAP data modeling framework
ODP	Operational Data Provisioning – SAP delta extraction framework
SLT	SAP Landscape Transformation – trigger-based real-time replication
OPDG	On-Premises Data Gateway – Microsoft gateway for on-prem data
NCo	SAP .NET Connector – library for SAP RFC communication
RFC	Remote Function Call – SAP native inter-system protocol
Direct Lake	Power BI mode reading Delta files directly from OneLake
DQ	DirectQuery – live query mode in Power BI